

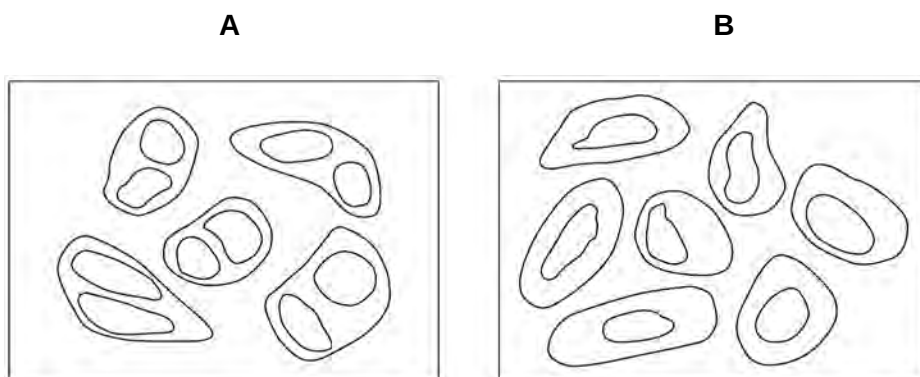
0 8 . 1

Scientists investigated a drug called MiTMAB as a treatment for cancer. MiTMAB inhibits cytokinesis.

Figure 8 shows drawings of cancer cells seen with an optical microscope from a:

- sample treated with MiTMAB
- control sample.

Figure 8



The cells in drawing **A** can be identified as those treated with MiTMAB.

Explain why.

[2 marks]

08.2

MiTMAB acts as a non-competitive inhibitor of an enzyme called dynamin.

Suggest how MiTMAB can cause dynamin to become inactive.

[3 marks]

Question 8 continues on the next page

When active, dynamin has **two** functions:

- it stimulates cytokinesis
- it inhibits cell death.

The scientists treated actively growing cultures of cancer cells with MiTMAB.

They incubated:

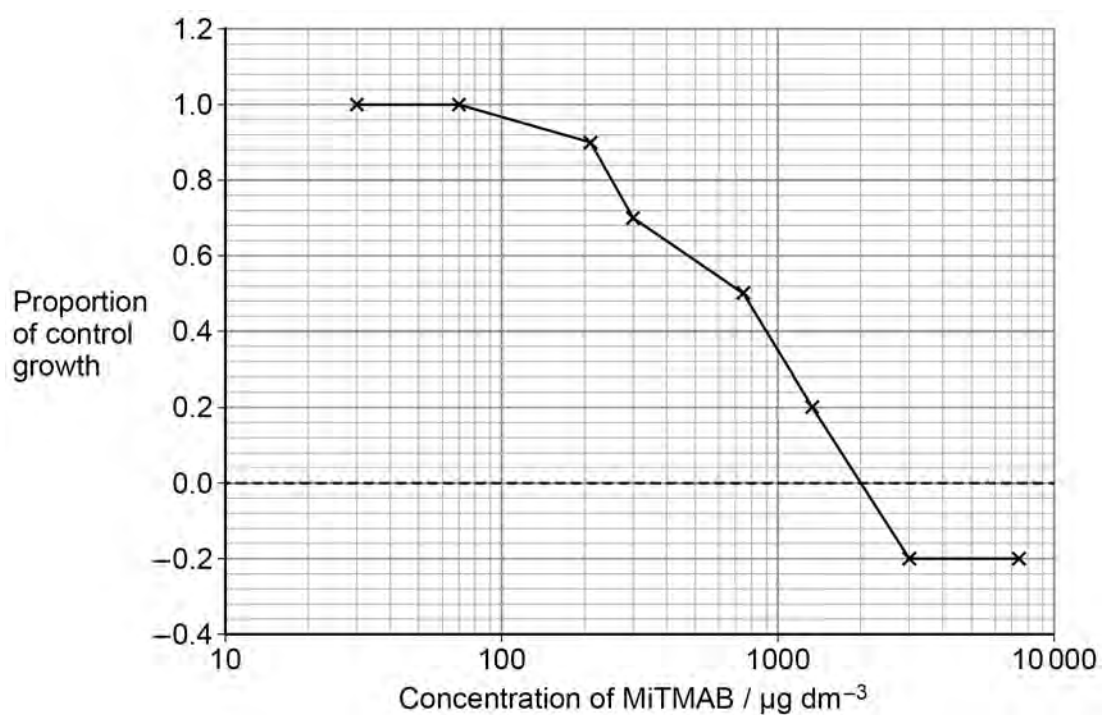
- one sample of 2500 cells without MiTMAB as a control
- eight samples, each with 2500 cells and a different concentration of MiTMAB.

After 72 hours, the scientists measured the number of cells in each sample.

Figure 9 shows the scientists' results.

A negative value for proportion of control growth means that fewer than 2500 cells were counted after 72 hours.

Figure 9



08.3

Use all the information given to explain the results shown in **Figure 9**.**[3 marks]**

08.4

0.01 dm³ of MiTMAB solution was added to the treated cells.

Calculate the increase in mass of MiTMAB (in µg) added to the cells to reduce the cell growth from equal to the control to 0.0 of the control.

Show your working.

[2 marks]

Answer _____ µg

1

Turn over for the next question**Turn over ►**

